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Read, Write, Execute

Read, Write, Execute: in computing, these are called permissions. Functionally speaking, they determine the extent of your authority within a computer or computer network, defining what exactly you can and cannot do. The right to *read* a file allows you to access its contents, while the right to *write* a file allows you to modify it. *Execution*, meanwhile, means that you have the ability to run a file or program, to carry out the actions it was designed to do.

Read, Write, Execute: this was my simple three-step plan. I wanted to burrow into the heart of the world's most secure network to find the truth, make a copy of it, and get it out into the world. And I had to do all this without getting caught—without being read, written, and executed myself.

Almost everything you do on a computer, on any device, leaves a record. Nowhere is this more true than at the NSA. Each log-in and log-out creates a log entry. Each permission I used left its own forensic trace. Every time I opened a file, every time I copied a file, that action was recorded. Every time I downloaded, moved, or deleted a file, that was recorded, too, and security logs were updated to reflect the event. There were network flow records, public key infrastructure records—people even joked about cameras hidden in the bathrooms, in the bathroom stalls. The agency had a not inconsiderable number of counterintelligence programs spying on the people who were spying on people, and if even one caught me doing something I wasn't supposed to be doing, it wouldn't be a file that was getting deleted.

Luckily, the strength of these systems was also their weakness: their complexity meant that not even the people running them necessarily knew how they worked. Nobody actually understood where they overlapped and where their gaps were. Nobody, that is, except the systems administrators. After all, those sophisticated monitoring systems you're imagining, the ones with scary names like MIDNIGHTRIDER—somebody's got to install them in the first